Lower bounds for time-dependent layout measurements

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Abstract

We prove lower bounds on the accounting exponent of OpenAI's layout interface for bounded-incidence, balanced-cut expanding scopes. The proof couples arbitrary time-dependent measurement spaces through a determinant. Under a capacity condition the charged exponent is at least a constant times $(\log\log n)^{3/2}/\sqrt{\log n}$. We also prove a small-measurement bound in auxiliary coordinates and a separate partner-time inequality. When the degree R is close to the separation Δ, the partner times cannot be dropped. Clifford-algebra polynomial matrices refute the far-pair matrix bound conjectured in the first (Zenodo) version of this paper, and make the measured rank an arbitrarily small fraction of $m(\Delta - R)$, even over the rationals. This needs $\Delta/R \to 1$. When Δ is at least a fixed multiple of R the far-pair bound remains open, and it would give a capacity-free bound of order $1/\sqrt{\log n}$. These are not general algorithmic running-time lower bounds.

1 Introduction

OpenAI [1] and Claude (Anthropic) [2] independently refute the exponential time hypothesis, with counting algorithms of running time $2^{n/(\log n)^{1/2+o(1)}}$ and $2^{O(n\log\log n/\sqrt{\log n})}$ respectively. We ask whether the square-root logarithm is forced by the accounting for OpenAI's layout interface, rather than by the problem itself.

Tulsyan [3] proves degree barriers for the fixed-M_0 template, in which M_0 has the prescribed Hamming support and fixes the kernel of a common measurement matrix. On the full label space, his first bound yields $R \geq (1 - o(1))\sqrt{Q\ln(1/\varepsilon)}$ in the even-alphabet moderate-deviation regime; his second is capacity-free and uses the Arora–Rao–Vazirani theorem [7]. He asks how to couple corrections that vary with time. Olivier [4] independently proves a degree lower bound for each fixed matrix satisfying OpenAI's equation (4.4). Neither fixed-M_0 result settles general time-dependent layouts.

Our determinant inequality couples all evaluation points without identifying their correction spaces. On expanding scopes, it yields lower bounds on the layout accounting exponent, and on the full auxiliary label space it gives a small-measurement degree bound. A separate interpolation argument proves a partner-time inequality and an exact result for a family of binary coset instances. Neither argument establishes a capacity-free barrier for general layouts. The second argument needs its partner times. When each direction has only one symmetric time, the far-pair matrix bound conjectured in our first version fails (Theorem 7), but only with R close to Δ. The slack case $\Delta \geq CR$, which is all the separation argument needs, remains open (Section 8).

2 The layout interface

Let $[n]$ be the variables and $I_f \subseteq [n]$ the scope of each factor f. OpenAI's layout has integers $r \geq 2$, $D \geq 1$, $1 \leq R < Q$ and $0 < \tau \leq 1/2$. For each $t \in [Q]$ it specifies a partition $[n] = \bigsqcup_{g=1}^{r} S_{t,g}$, empty parts permitted, with $|S_{t,g}| \leq 2n/r$. Write $a_v(t) = g$ when $v \in S_{t,g}$, and write $d_f(t)$ for the assigned group of factor f.

For each (t, g) there is an integer matrix $C_{t,g}$ on columns $S_{t,g}$, with $c_{t,g}$ rows, each of ℓ_1 norm at most D, and $\sum_g c_{t,g} \leq 2\tau n$. For each incidence $v \in I_f$ there is a row $P_{f,v}(T) \in \mathbb{Q}[T]^{1\times n}$ of degree at most R with $P_{f,v}(0) = e_v$. For every $g \neq d_f(t)$ and $\xi \in \ker C_{t,g}$, the row $P_{f,v}(t)$ annihilates the extension of ξ by zero. Equivalently, its restriction to $S_{t,g}$ belongs to the row space of $C_{t,g}$. Finally, with $d_{\mathrm{den}} = (R-1)!Q!$, each coefficient of $d_{\mathrm{den}}P_{f,v}$ is an integer of $O(Q \log Q + \log D)$ bits, with an absolute constant. Our lower bounds do not use this coefficient condition.

For total product degree $d = O(n)$, the counting analysis of [1] gives the following upper-bound proxy for the normalized exponential cost, apart from polynomial factors:

$$F = \frac{R}{Q} \log_2 \frac{2Q}{R} + \tau \log_2(2D+1) + \frac{1}{r}. \tag{1}$$

A lower bound on F limits this particular accounting; it is not a lower bound on the running time of every algorithm, or even on an alternative implementation of the counting step.

3 A determinant inequality

The following result uses only the pointwise row-space condition, not expansion or integrality. It holds over any field $\mathbb{F}$ of characteristic zero, with distinct nonzero evaluation points t; signs are ordinary real signs applied to schedules.

Theorem 1. *Choose signs $\epsilon_{t,g} \in \{-1, 1\}$ and put $s(w) = \sum_t \epsilon_{t,w(t)}$. Suppose m distinct variables v_i have incident factors f_i with $s(a_{v_i}) \leq -L$ and $s(d_{f_i}) \geq L$. If $c_t = \sum_g c_{t,g}$, then*

$$m(L - R) \leq \sum_{t=1}^{Q} c_t.$$

The factors may repeat and the measurement spaces may vary arbitrarily with t.

Proof. Let $A(T)_{ij} = P_{f_i,v_i}(T)_{v_j}$. Then $A(0) = I_m$, so $\det A$ is nonzero and has degree at most mR. At t, let $u_{t,g}$ and $v_{t,g}$ count the selected rows assigned to g and selected columns in g. Form M_t by retaining entries of $A(t)$ only where row and column groups differ. In each column group its rows lie in the restricted row space of $C_{t,g}$, so $\operatorname{rank} M_t \leq c_t$. The remainder has matching-group blocks, hence

$$\operatorname{rank} A(t) \leq c_t + \sum_g \min(u_{t,g}, v_{t,g}) = m + c_t - \Delta_t, \quad \Delta_t = \tfrac{1}{2} \sum_g |u_{t,g} - v_{t,g}|.$$

If a polynomial matrix has rank ρ at t, constant invertible row operations make $m - \rho$ rows vanish there; factoring $T - t$ from each proves that $(T - t)^{m-\rho}$ divides its determinant. This elementary multiplicity fact is also a consequence of Smith form [5]. Summing over distinct points,

$$mR \geq \sum_t (m - \operatorname{rank} A(t)) \geq \sum_t \Delta_t - \sum_t c_t.$$

Finally $2mL \leq \sum_{t,g} \epsilon_{t,g}(u_{t,g} - v_{t,g}) \leq 2\sum_t \Delta_t$. Combining the inequalities proves the claim. □

4 Expanding scopes

Assume each variable occurs in at most b scopes and every $W \subseteq [n]$ with $n/4 \le |W| \le n/2$ is crossed by at least $h|W|$ factors, where a crossing factor meets both W and its complement. Here b and $h > 0$ are fixed; these are additional hypotheses, not part of the layout definition. Set $\alpha = h/(8b)$.

Lemma 2. *For $Q \ge (512/\alpha)^2$, $0 \le L \le \alpha Q/64$, and two schedules at Hamming distance at least αQ, independent uniform signs $\epsilon_{t,g}$ satisfy*

$$\Pr[s(a) \le -L,\ s(d) \ge L] \ge c_\alpha \exp[-C_\alpha(L^2/Q + 1)]$$

for positive constants depending only on α.

Proof. Let $\ell \ge \alpha Q$ be the distance. At the ℓ mismatched times the two signs $\epsilon_{t,a(t)}$ and $\epsilon_{t,d(t)}$ are independent; let k count those where they disagree. Then $k \sim \mathrm{Bin}(\ell, 1/2)$ and Paley–Zygmund gives $\Pr(k \ge \ell/4) \ge 1/8$. Conditional on these positions, $s(a) = C - Z$ and $s(d) = C + Z$, where C and Z are independent sums of $Q-k$ and k fair signs. Chebyshev gives $\Pr(|C| \le 2\sqrt{Q}) \ge 3/4$. Put $u = L + 2\sqrt{Q}$ and $w = u + 2\sqrt{k}$. If $w \le k/4$, exponential tilting with $k \tanh\lambda = w$ and Chebyshev under the tilted law yield

$$\Pr(Z \ge u) \ge \tfrac{3}{4} \exp\{-3w(u + 4\sqrt{k})/k\}.$$

Indeed $\lambda \le 3w/k$, and the tilted probability of $|Z - w| < 2\sqrt{k}$ is at least $3/4$; undo the tilt using $(\cosh\lambda)^k \ge 1$. The interval argument needs no parity adjustment. On $k \ge \alpha Q/4$, the stated range gives $w \le L + 4\sqrt{Q} \le \alpha Q/64 + \alpha Q/128 < \alpha Q/16 \le k/4$. On the event $|C| \le 2\sqrt{Q}$ and $Z \ge u$ we have $s(d) = C + Z \ge L$ and $s(a) = C - Z \le -L$. The exponent is $O_\alpha(L^2/Q + 1)$, and multiplying the three probability bounds proves the lemma. □

Theorem 3. *Under the above expansion hypotheses, every layout satisfies, for a constant $c > 0$ depending only on b, h,*

$$F \ge \frac{c}{\sqrt{Q}}, \qquad F \ge c\frac{\log_2(2Q)}{\sqrt{Q}}.$$

If also $Q \log_2 r \le K \log_2 n$ for a fixed K, then for sufficiently large n,

$$F \ge c_{b,h,K} \frac{(\log_2 \log_2 n)^{3/2}}{\sqrt{\log_2 n}}.$$

Proof. For $r \ge 8$, partition parts can be greedily united to give $n/4 \le |W| \le n/2$. Each crossing factor has an incidence whose factor and variable groups differ, so there are at least $hn/4$ such incidences at each time. There are at most bn incidences, hence at least $hn/8$ have schedule distance at least αQ. Lemma 2, expectation, and selection of one incidence per variable give signs and $m \ge cn \exp(-CL^2/Q)$ selected variables. Theorem 1 and $c_t \le 2\tau n$ imply, whenever $L > R$ is in the lemma's range,

$$\tau \ge c\frac{L - R}{Q} \exp(-CL^2/Q). \tag{2}$$

For $R \le \sqrt{Q}$, take $L = R + \lceil\sqrt{Q}\rceil$; for the lemma's lower bound on Q, this is admissible and gives $\tau \ge c/\sqrt{Q}$, hence $F \ge \tau \log_2 3 \ge c/\sqrt{Q}$ since $D \ge 1$. For $R > \sqrt{Q}$, the first term of (1) is at least $R/Q > 1/\sqrt{Q}$. Bounded Q and $r < 8$ are handled by reducing c, using respectively $R/Q \ge 1/Q$ and $1/r > 1/8$.

For the stronger bound, integer measurement rows have support at most D. At each time the union of their supports has size at most $2\tau nD$. An off-assigned incidence (f, v) outside this union forces $P_{f,v}(t)_v = 0$; its diagonal polynomial is nonzero at zero and has at most R roots. Summing over at most bn incidences gives

$$R/Q + 2\tau D \geq h/(4b). \tag{3}$$

If $R > \sqrt{Q}$, concavity of $x \log_2(2/x)$ on $[Q^{-1/2}, 1]$ gives the claimed bound. Otherwise, for large Q, (3) gives $D \geq h/(16b\tau)$ and the preceding argument gives $\tau \geq c_0/\sqrt{Q}$. The function $\tau \log_2(1+z/\tau)$ is increasing for $z > 0$, so the measurement term is $\Omega(\log(2Q)/\sqrt{Q})$. Again bounded Q and $r < 8$ are absorbed into the constant.

For the capacity conclusion put $N = \log_2 n$. If $r \leq N^{1/4}$, then $1/r \geq N^{-1/4}$ dominates the claimed bound. Otherwise $Q \leq 4KN/\log_2 N$. The function $\log_2(2Q)/\sqrt{Q}$ decreases for sufficiently large Q; evaluation at that upper endpoint proves the result, while bounded Q gives a constant. □

5 Auxiliary coordinates

Let B be a nonempty finite label set. For each $u \in B$ let $a_u, d_u \in [r]^Q$, and let $P_u(T) \in \mathbb{Q}[T]^B$ have degree at most R and $P_u(0) = e_u$. At time t let $\Lambda_t \subseteq \mathbb{Q}^B$ have dimension at most $\varepsilon|B|$, and suppose $P_u(t) - \lambda_{u,t}$ is supported on $\{w : a_w(t) = d_u(t)\}$ for some $\lambda_{u,t} \in \Lambda_t$. Corrections may depend on both u and t.

Theorem 4. *Fix $\alpha, \beta, K > 0$. If at least $\beta|B|$ labels satisfy $\operatorname{dist}(a_u, d_u) \geq \alpha Q$ and $0 < \varepsilon \leq K/Q$, then, for sufficiently large Q,*

$$R \geq c \min\{Q, \sqrt{Q \ln(1/\varepsilon)}\},$$

where $c > 0$ depends only on α, β, K. In particular this holds on the full label space $B = ([r] \times [r])^Q$ for $r \geq 2$, with constants independent of r.

Proof. Lemma 2 selects $m \geq c|B|e^{-CL^2/Q}$ long labels. The determinant proof applies with the whole correction matrix of rank at most $\varepsilon|B|$ at each time, yielding

$$\varepsilon \geq c(L - R)Q^{-1}e^{-CL^2/Q}.$$

If $R < \sqrt{Q}$, taking $L = R + \lceil\sqrt{Q}\rceil$ contradicts $\varepsilon \leq K/Q$ for large Q. If $2R > \alpha Q/64$, the linear branch of the conclusion holds. Otherwise take $L = 2R$: the inequality gives $\varepsilon \geq (c/\sqrt{Q})e^{-4CR^2/Q}$. As $\ln(1/\varepsilon) \geq \ln Q - \ln K$, rearrangement gives $R^2 \geq cQ \ln(1/\varepsilon)$ for large Q. On the full space the mean distance is $(1 - 1/r)Q \geq Q/2$, so at least one third of labels have distance at least $Q/4$. □

The long-pair assumption cannot be dropped: even with balanced schedules, take $B = [r]^Q$, $a_u = d_u = u$, $P_u = e_u$, $R = 1$, and one-dimensional correction spaces (or zero corrections). Then $\varepsilon = 1/|B|$ is allowed and no such degree bound follows. In OpenAI's incidence embedding, $\tau = rb\varepsilon$ is a sufficient physical budget, not necessarily the smallest possible budget; the auxiliary theorem itself incurs no $\ln(rb)$ loss.

6 Partner times

Let $A(T)$ be an $m \times m$ polynomial matrix of degree at most R with $A(0) = I$. All evaluation times are distinct and nonzero. At time t, rows and columns have group labels $d_i(t), a_j(t)$. Let $N_t = \{(i, j) : d_i(t) = a_j(t)\}$ be the unmeasured entries, and let $I_t = \{i : d_i(t) \neq a_i(t)\}$. Let cost_t be the sum, over column groups, of the ranks of the measured (off-N_t) column blocks. This is at most the number of layout measurement rows when A is a selected incidence matrix.

Proposition 5. *For each t in a set $\mathcal{T}$, suppose there is a set $P(t)$ of R distinct nonzero times, excluding t, at which every entry of N_t is measured. For $t' \in P(t)$ let $\beta(t,t')$ be the maximum, over column groups g' at t', of the number of labels h for which some column has $a_j(t) = h$ and $a_j(t') = g'$. Put*

$$\lambda = \max_{t'} \sum_{t \in \mathcal{T} : t' \in P(t)} \beta(t,t').$$

Then, with cost summed over all evaluation times,

$$(1+\lambda) \sum_t \mathrm{cost}_t \geq \sum_{t \in \mathcal{T}} |I_t| - mR.$$

Proof. Let $G_t(T)$ retain the entries of $A(T)$ in N_t with row index in I_t, and zero all other entries. These entries are off-diagonal, so $G_t(0) = 0$ and $\deg G_t \leq R$. At t, the measured part has rank at most cost_t, while the unmeasured part in rows outside I_t has rank at most $m - |I_t|$. Thus the corank of $A(t)$ is at least $|I_t| - \mathrm{cost}_t - \operatorname{rank} G_t(t)$.

At a partner t', within each column group g', $G_t(t')$ splits into at most $\beta(t,t')$ rectangles, indexed by h, with rows $i \in I_t$, $d_i(t) = h$, and columns $a_j(t) = h$, $a_j(t') = g'$. Each is a row-and-column restriction of the measured block at t'. Hence $\operatorname{rank} G_t(t') \leq \beta(t,t')\mathrm{cost}_{t'}$. Lagrange interpolation through 0 and $P(t)$ gives $\operatorname{rank} G_t(t) \leq \sum_{t' \in P(t)} \beta(t,t')\mathrm{cost}_{t'}$. Summing the corank inequalities and using the determinant multiplicity bound $\sum_t \operatorname{corank} A(t) \leq mR$ gives the result. □

Corollary 6. *Let indices be $x \in \mathbb{F}_2^k$, $m = 2^k$. For each of Δ nonzero directions y_ℓ, take one antisymmetric time with $a_x = \langle x, y_\ell \rangle$, $d_x = 1 - a_x$, and R symmetric times with $d_x = a_x$ using the same direction. Then every such A has total cost at least $m(\Delta - R)$; for $R \leq \Delta$ this is attained.*

Proof. At an antisymmetric time the measured part is block diagonal and the unmeasured part is block antidiagonal. Write G_y for the block-antidiagonal projection. The determinant bound gives $\sum_{t \,\mathrm{anti}} \mathrm{cost}_t \geq m(\Delta - R) - \sum_{t \,\mathrm{anti}} \operatorname{rank} G_{y_t}(A(t))$. Each $G_y(A(T))$ vanishes at zero; interpolation through its R symmetric partners bounds its rank at the antisymmetric time by the sum of their measured-block ranks. The partner sets are disjoint, giving the bound. For $R \leq \Delta$, choose a scalar polynomial p of degree R, with $p(0) = 1$ and roots at any R antisymmetric times, and take $A(T) = p(T)I$: its total cost is exactly $m(\Delta - R)$. At symmetric times the unmeasured part is block diagonal and contains the identity. Consequently the determinant rank estimate of Theorem 1 alone charges nothing there; interpolation brings those times into the argument. □

In the family of Corollary 6, a row x and a column x' differ at the antisymmetric time of direction y_ℓ when $\langle x, y_\ell \rangle = \langle x', y_\ell \rangle$ and at its R symmetric times otherwise, so $\mathrm{dist}(d_x, a_{x'}) \geq \Delta$ for every pair. The first public version of this paper (Zenodo, version 1.0) conjectured that separation alone suffices: there is an absolute $c > 0$ such that, for arbitrary row and column schedules with $\mathrm{dist}(d_i, a_j) \geq \Delta > R$ for all $i \leq j$, every degree-R matrix A with $A(0) = I$ has total measured-block rank at least $c\,m(\Delta - R)$. In Tulsyan's second theorem, ARV [7] separates two large sets of variable schedules, and linkedness supplies many incidence labels with all selected row–column pairs far apart, so the all-pairs case of this bound would have extended that route to time-dependent measurements. The next section shows that the conjecture is false, already in the all-pairs case with two groups.

7 Partner times cannot be dropped when R is close to Δ

The *paired j-cube* has indices $x \in \mathbb{F}_2^j$, $m = 2^j$, two groups, and times $[Q]$ with $Q = 2j$. Direction $\ell \in \{0, \ldots, j-1\}$ has a symmetric time $\sigma_\ell = 2\ell + 1$, with $a_x = d_x = x_\ell$, and an antisymmetric time

$\alpha_\ell = 2\ell + 2$, with $a_x = x_\ell$, $d_x = 1 - x_\ell$. This is the family of Corollary 6 with $y_\ell = e_\ell$ and $\Delta = j$, but with one symmetric time per direction instead of R. The entry (x, y) is measured at α_ℓ if and only if $x_\ell = y_\ell$, and at σ_ℓ if and only if $x_\ell \neq y_\ell$. Hence $\operatorname{dist}(d_x, a_y) = j$ for every pair x, y, including $x = y$.

Theorem 7. *Let $q \geq 3$ and $s \geq 1$, and put $j = s(2q^2 + 2q + 2)$, $R = 2sq(q + 1)$ and $\Delta = j$, so that $\Delta - R = 2s$. On the paired j-cube there is a matrix $A(T)$ of degree at most R, with $A(0) = I$ and entries in a number field, whose total measured-block rank is*

$$(q + 1)\, 4^{-q}\, m(\Delta - R).$$

Repeating every row and column schedule e times gives a rational matrix with the same ratio. Hence the far-pair bound fails for every $c > 0$, over $\mathbb{Q}$, with two balanced groups, times $[Q]$, and $\operatorname{dist}(d_i, a_j) = \Delta$ for all pairs.

We build the matrices from Clifford algebras. Fix q directions P. For $w \in \mathbb{F}_2^q$ put $h_w(T) = \prod_{p \in P}(T - \rho_p(w))$, where $\rho_p(w) = \sigma_p$ if $w_p = 1$ and $\rho_p(w) = \alpha_p$ otherwise. For $(w, v) \in \mathbb{F}_2^q \times \mathbb{F}_2^q$ let $G_{w,v}$ be the $2^q \times 2^q$ signed permutation matrix with entry $(-1)^{v \cdot y}$ at $(y + w, y)$. Then $G_{w,v}^2 = (-1)^{w \cdot v} I$, $G_{w,v}$ has trace zero unless $(w, v) = 0$, and $G_{w,v}$, $G_{w',v'}$ anticommute if and only if $w \cdot v' + w' \cdot v = 1$. A *$q$-family* is a list of $2q + 1$ pairwise anticommuting labels (w, v) with distinct w, such that the $2q + 1$ polynomials h_w^2 are linearly independent, hence a basis of the polynomials of degree at most $2q$.

Lemma 8. *Given a q-family on P and distinct nonzero times $t_1, \ldots, t_{2q+2}$ outside $\{\sigma_p, \alpha_p : p \in P\}$, there is a $2^q \times 2^q$ matrix $M(T)$ of degree at most $q+1$ over a number field with $M(0) = I$, such that every entry measured at a time σ_p or α_p, $p \in P$, vanishes there,* $\det M = g^{2^{q-1}}$ *with $g = \prod_l (1 - T/t_l)$, and* $\operatorname{rank} M(t_l) = 2^{q-1}$ *for every l.*

Proof. Let $F = \prod_{p \in P}(1 - T/\alpha_p)$ and $H_w = T h_w$. Choose b so that $F^2(1 + bT)^2 - g$ has no linear term. That difference is then $-T^2$ times a polynomial of degree at most $2q$, so the basis property gives unique k_w with $g = F^2(1 + bT)^2 - \sum_w k_w H_w^2$. Choose c_w with $c_w^2 = (-1)^{w \cdot v} k_w$, and put $d = F(1 + bT)$, $N = \sum_w c_w H_w G_{w,v}$ and $M = dI + N$. Then $M(0) = I$ and $\deg M \leq q + 1$. Because the w are distinct, the entry (x, y) of N involves only the term with $w = x + y$. If it is measured at α_p, then $w_p = 0$, and both H_w and d (through F) vanish at α_p. If it is measured at σ_p, then $w_p = 1$ and $x \neq y$, and H_w vanishes at σ_p.

Anticommutation gives $N^2 = \kappa I$ with $\kappa = \sum_w k_w H_w^2 = d^2 - g$. The polynomial κ is nonzero, since otherwise $g = d^2$ would vanish at α_p. Over an algebraic closure of the field of rational functions, N is diagonalizable with eigenvalues $\pm\sqrt{\kappa}$, and both multiplicities are 2^{q-1} because N has trace zero. Hence $\det M = g^{2^{q-1}}$. At a root t_l of g we have $\kappa(t_l) = d(t_l)^2$. If $d(t_l) \neq 0$, then $M(t_l)^2 = 2d(t_l) M(t_l)$, so $M(t_l)/(2d(t_l))$ is an idempotent of trace 2^{q-1}. If $d(t_l) = 0$, then $M(t_l)^2 = 0$, so $\operatorname{rank} M(t_l) \leq 2^{q-1}$, while the nullity is at most the order 2^{q-1} of the root of $\det M$, since t_l is a simple root of g. □

Lemma 9. *For every $q \geq 3$ there is a q-family on any q consecutive directions.*

Proof. Translating all times by a constant replaces each $h_w(T)$ by $h_w(T - c)$, a unipotent change of basis, so linear independence does not depend on the placement. For $q = 3, 4, 5$ the families

$(1, 0), (2, 1), (3, 5), (4, 3), (5, 1), (6, 3), (7, 5),$
$(0, 3), (1, 0), (2, 1), (5, 1), (6, 11), (9, 5), (10, 3), (13, 5), (14, 11),$
$(5, 26), (6, 18), (7, 23), (10, 30), (14, 17), (16, 0), (17, 31), (19, 19), (22, 27), (24, 22), (28, 17)$

(bit i of each integer is direction i) satisfy the axioms, and the coefficient matrices of their polynomials h_w^2 have determinants 4096, 715456512 and 176287042228935720960 in exact arithmetic. For larger q, let Γ_1 be a q_1-family on P_1 and Γ_2 a q_2-family on disjoint P_2, and fix $\gamma^* \in \Gamma_1$. The labels $\gamma \otimes I$ for $\gamma \in \Gamma_1 \setminus \{\gamma^*\}$ and $\gamma^* \otimes \delta$ for $\delta \in \Gamma_2$ number $2(q_1+q_2)+1$. They anticommute pairwise, because the symplectic form adds over tensor factors, and their X-parts $(w_\gamma, 0)$ and (w^*, w_δ) are distinct. Their polynomials are $h_{w_\gamma}^{(1)2} F_2'^2$ and $e\, h_{w_\delta}^{(2)2}$, where $F_2' = \prod_{p\in P_2}(T-\alpha_p)$ and $e = h_{w^*}^{(1)2}$. A linear relation reads $AF_2'^2 = -eB$, with A in the span of the first kind and B in the span of the second. The roots of e are times of P_1 and those of F_2' are times of P_2, so e divides A. Since $\deg A \le 2q_1 = \deg e$, $A = \lambda e$, and independence of Γ_1 forces $\lambda = 0$ and all coefficients of A to vanish. Then $B = 0$, and independence of Γ_2 finishes the proof. Every $q \ge 6$ is a sum of 3s and 4s. □

Proof of Theorem 7. Put $h = 2q+2$ and $k = 2q$. Split the directions into s groups, each consisting of k blocks of q consecutive directions and h *root directions*. Within a group, let every block carry the matrix of Lemma 8 (with Lemma 9) whose roots are the times α_ℓ of that group's root directions. Let A be the Kronecker product of these sk matrices and of the identity on the root bits. Then $A(0) = I$ and $\deg A \le sk(q+1) = R$. An entry of A measured at a time of a block direction is a product that contains a measured entry of that block's factor, so it is zero. At σ_ℓ for a root direction ℓ, the measured entries have $x_\ell \ne y_\ell$, where A vanishes. At α_ℓ the measured part consists of the two diagonal blocks $x_\ell = y_\ell$, so the cost there is $\operatorname{rank} A(\alpha_\ell) = m \prod_i \operatorname{rank} M_i(\alpha_\ell)/2^q$. The k factors of the group containing ℓ have rank 2^{q-1} there, and all other factors are invertible. The total cost is therefore $m \cdot sh \cdot 2^{-k}$, while $\Delta - R = s(kq+h) - sk(q+1) = 2s$, which gives the ratio $(q+1)4^{-q}$.

For the rational version, let K be the number field generated by the entries of A, of degree e over $\mathbb{Q}$, and replace each entry by its $e \times e$ regular representation. Copy every row and column schedule e times. Measured index sets are unions of whole copy-blocks, the regular representation of a K-matrix has $\mathbb{Q}$-rank e times its K-rank, and m also grows by the factor e. Distances and degree are unchanged. □

Within its own class the construction is optimal. With c_ℓ factors sharing the root direction ℓ, convexity of 2^{-c} shows that $\sum_\ell 2^{-c_\ell} \ge (q+1)4^{-q}(\Delta - R)$ for factors of size at most q. In terms of the degree alone ($s = 1$), its cost is $m(\Delta - R)\, 2^{-(1+o(1))\sqrt{2R}}$.

The saving needs $\Delta/R \to 1$: here $\Delta - R = 2s$ while $R = 2sq(q+1)$. Within the class this is unavoidable, even with clusters of different sizes. Suppose clusters of sizes q_i and s_0 scalar roots use total degree $\sum_i (q_i+1) + s_0 \le R$. Then at least $\Delta - R$ root directions escape the scalar roots, and they carry at most $2R$ cluster roots. Convexity of 2^{-c}, and $2^{-c} \ge 1 - c/2$ for integers $c \ge 0$, bound the cost below by

$$m \max\{(\Delta - R)\, 2^{-2R/(\Delta - R)},\ \Delta - 2R\}.$$

So for $\Delta \ge CR$ with a fixed $C > 1$ the class saves at most a constant factor. Much smaller instances already beat $m(\Delta - R)$. On the paired 4-cube with $R = 2$, an explicit real matrix has cost $24 < 32$. It has $m = 16$, beyond the $m \le 12$ reach of our earlier searches. The factorization $(dI + N)(dI - N) = gI$ is the norm form of a Clifford algebra, as in matrix factorizations [6]. Theorem 7 concerns the abstract matrix statement only. It does not construct a layout, and it says nothing about coefficient sizes, measurement-row norms or linked expanding scopes. We do not know whether those conditions, enough partner times as in Corollary 6, or slack $\Delta \ge CR$ restore a bound of order $m(\Delta - R)$.

8 Limitations and verification

It remains open whether, on the linked bounded-incidence scope systems used in Tulsyan's second theorem, sufficiently small time-dependent measurement budgets force $R = \Omega(Q/\sqrt{\log n})$ without a capacity assumption. A sufficient matrix statement is the *slack far-pair bound*: for some constants $C > 1$ and $c > 0$, if $\operatorname{dist}(d_i, a_j) \geq \Delta \geq CR$ for all pairs, then every degree-R matrix A with $A(0) = I$ has total measured-block rank at least $cm\Delta$. In Tulsyan's second theorem with $r \geq 3$, ARV and linkedness give $m = \Omega(n)$ incidences with all selected pairs at distance $\Delta = \Omega(Q/\sqrt{\log n})$. If $R \leq \Delta/C$, the bound and $c_t \leq 2\tau n$ give $\tau = \Omega(1/\sqrt{\log n})$; otherwise $R/Q = \Omega(1/\sqrt{\log n})$. Either way $F = \Omega(1/\sqrt{\log n})$, with no capacity assumption, and $r \leq 2$ gives $F \geq 1/2$. Theorem 7 does not refute the slack bound, because it needs $\Delta/R \to 1$. A simple equal-marginal example has row schedules $00, 11$ and column schedules $01, 10$: every coordinate imbalance is zero although all cross-pairs are separated. It shows why the additive determinant charge does not directly capture the nonlinear separation used in the fixed-correction argument.

On the paired cube of Section 7, the bound holds whenever the off-diagonal entries have bounded degree, with or without slack.

Proposition 10. *On the paired j-cube, let $A(T)$ have degree at most R with $A(0) = I$, and let its off-diagonal entries have degree at most an integer $d \geq 1$. Then the total measured-block rank is at least $m(j - R)/(d\,2^{d-1})$. In particular it is at least $m\Delta/(d\,2^d)$ when $\Delta = j \geq 2R$.*

Proof. Let Z_ℓ be the diagonal matrix with entries $(-1)^{x_\ell}$, and put $P_\ell^{\pm} X = (X \pm Z_\ell X Z_\ell)/2$. These projections commute. The measured part is $P_\ell^{+} A(\alpha_\ell)$ at α_ℓ and $P_\ell^{-} A(\sigma_\ell)$ at σ_ℓ. Let r_ℓ and s_ℓ be their ranks, and $C = \sum_\ell (r_\ell + s_\ell)$. By the determinant multiplicity bound,

$$m(j - R) \leq \sum_\ell \operatorname{rank} A(\alpha_\ell) \leq \sum_\ell r_\ell + \sum_\ell \operatorname{rank} P_\ell^{-} A(\alpha_\ell).$$

The claim is immediate if $j \leq R$. Otherwise the off-diagonal entries have degree at most $\min\{d, R\}$, and $u\,2^{u-1}$ increases with u, so we may assume $1 \leq d \leq R < j$; in particular $j \geq 2$. Fix ℓ and a set H of $d - 1$ other directions. The entries of $P_\ell^{-} A$ are off-diagonal entries of A, which vanish at 0, so $B = P_\ell^{-} A/T$ is a polynomial of degree at most $d - 1$. For $\varepsilon \in \{\pm\}^H$ put $Q_\varepsilon = \prod_{h \in H} P_h^{\varepsilon_h}$, so that $\sum_\varepsilon Q_\varepsilon$ is the identity. Let $t_h = \alpha_h$ if $\varepsilon_h = +$ and $t_h = \sigma_h$ otherwise. Interpolate $Q_\varepsilon B$ at the d distinct nonzero nodes σ_ℓ and t_h ($h \in H$), evaluate at α_ℓ, and sum over ε. This writes $P_\ell^{-} A(\alpha_\ell)$ as a combination of the terms $Q_\varepsilon P_\ell^{-} A(\sigma_\ell)$ and

$$Q_\varepsilon P_\ell^{-} A(t_h) = P_\ell^{-} \textstyle\prod_{k \in H \setminus \{h\}} P_k^{\varepsilon_k} \big(P_h^{\varepsilon_h} A(t_h)\big),$$

where $P_h^{\varepsilon_h} A(t_h)$ is the measured part at t_h. Expand each projection into conjugations by products of the Z_k. The σ_ℓ data then enter through at most 2^{d-1} conjugates of $P_\ell^{-} A(\sigma_\ell)$. For each $h \in H$ and each sign, the measured part at t_h enters through at most 2^{d-1} conjugates. Hence

$$\operatorname{rank} P_\ell^{-} A(\alpha_\ell) \leq 2^{d-1} \Big(s_\ell + \sum_{h \in H} (r_h + s_h) \Big).$$

Average over all $(d-1)$-sets H of directions other than ℓ, so that each such direction has weight $(d-1)/(j-1)$, and sum over ℓ. This gives $\sum_\ell \operatorname{rank} P_\ell^{-} A(\alpha_\ell) \leq 2^{d-1} \big(\sum_\ell s_\ell + (d-1)C \big)$. Therefore $m(j - R) \leq \big(1 + (d-1)2^{d-1}\big) \sum_\ell r_\ell + d\,2^{d-1} \sum_\ell s_\ell \leq d\,2^{d-1} C$. □

For quadratic and cubic off-diagonal parts, the proposition gives $m\Delta/8$ and $m\Delta/24$ at $\Delta \geq 2R$. The diagonal part may have any degree up to R. The interpolation uses the measured data of other

directions, not only the partner times of direction ℓ. The loss $d\,2^{d-1}$ is exponential, however. The matrices of Theorem 7 have off-diagonal entries of degree close to R, so the proposition says little about them. For each fixed d this proves the slack bound with a constant depending on d. It gives no constant uniform in d when the off-diagonal degree grows with R, which is the open case on the paired cube.

Even with the capacity condition, our bound is smaller by a factor $\sqrt{\log\log n}$ than the order obtained in [1], namely $(\log\log n)^2/\sqrt{\log n}$. This cannot be closed by optimizing (2) and combining it only with (3). Put $N = \log_2 n$ and take

$$Q = \lfloor N/\log_2 N \rfloor, \quad R = 1, \quad r = \lceil \sqrt{Q} \rceil, \quad \tau = \kappa_1/\sqrt{Q}, \quad D = \lceil \kappa_2 \sqrt{Q} \rceil,$$

with sufficiently large constants κ_1, κ_2. These formal parameters satisfy the displayed necessary inequalities and $Q \log_2 r = (1/2 + o(1))N$, while $F = \Theta(\log Q/\sqrt{Q})$. They do not construct a layout: additional polynomial and incidence constraints may exclude them. The capacity condition itself is not part of the layout definition.

The repository's exact certificate checker verifies finite layouts and determinant certificates, including equality on a four-cycle and genuinely varying measurement spaces on the four-vertex complete graph. It checks a fixed-schedule feasibility claim and small sign-mixture probabilities; the latter do not establish the asymptotic probability constants. The partner checker verifies a projection identity and the corresponding block-rank identity exactly. The far-pair checker verifies the base families of Lemma 9, the recursive families up to $q = 10$, and the matrices of Lemma 8 for $q = 3, 4, 5$, all in exact arithmetic. The slack checker verifies the interpolation identity in the proof of Proposition 10 exactly, on the paired 4-cube for $d = 1, 2, 3$. These finite calculations do not replace the general proofs.

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